

How to identify Source free response and Forced response situation

For $t > 0$, if the inductor/capacitor in the circuit **NOT** connected to any **DC source**, then determine **SOURCE FREE response**.

$$\text{Current through inductor: } i_L(t) = i_L(0)e^{-\frac{t}{\tau}}, \text{ for } t > 0$$

$$\text{Voltage across capacitor: } V_C(t) = V_C(0)e^{-\frac{t}{\tau}}, \text{ for } t > 0$$

For $t > 0$, if the inductor/capacitor in the circuit is **connected to DC sources**, then determine **Forced response**.

$$\text{Current through inductor: } i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]e^{-\frac{t}{\tau}}, \text{ for } t > 0$$

$$\text{Voltage across capacitor: } V_C(t) = V_C(\infty) + [V_C(0) - V_C(\infty)]e^{-\frac{t}{\tau}}, \text{ for } t > 0$$

$i_L(\infty)$ and $V_C(\infty) \rightarrow \text{Solve the circuit for } t > 0$

$i_L(0)$ and $V_C(0) \rightarrow \text{Solve the circuit for } t < 0$

$R_{TH} \rightarrow \text{Solve the circuit for } t > 0, \text{ open the capacitor or inductor and look through the nodes to find } R_{TH}$